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journal homepage: <http://ees.elsevier.com/apjtm>Original research <http://dx.doi.org/10.1016/j.apjtm.2015.05.010>Antitumor activity of *Aponogeton undulatus* against Ehrlich ascites carcinoma in Swiss albino miceMd. Reyazul Islam^{1*,5}, Md. Badrul Alam^{2,4,5}, Umme Tamima³, Shayla Islam Jenny⁴¹Department of Bioactive Materials Science, Chonbuk National University, 567, Baekje-daero, Deokjin-gu, Jeonju-si, Jeollabuk-do, Republic of Korea²Graduate School of Food Science & Biotechnology Kyungpook National University, 80, Daehak-ro, Buk-gu, Daegu, Republic of Korea³College of Pharmacy, Seoul National University, 1 Gwanak-ro, Gwanak-gu, Seoul, Republic of Korea⁴Department of Pharmacy, AtishDipankar University of Science & Technology, Dhaka, Bangladesh

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ABSTRACT

Objective: To investigate *in vitro* antioxidant and *in vivo* antitumor activity of the crude methanolic extract of *Aponogeton undulatus* (A. *undulatus*) (MAU) along with its various organic fractions.**Methods:** A. *undulatus* leaves were successively extracted using methanol (MAU) and then fractionated by chloroform, ethyl acetate (EAU) and water. The total antioxidant capacity, lipid peroxidation inhibition assay, 1, 1-diphenyl-2-picrylhydrazyl (DPPH) free radical scavenging assay and ferrous reducing power assessment were used to evaluate the antioxidant potential of the crude extract and its organic fractions. The *in vivo* antitumor activity is evaluated against Ehrlich ascites carcinoma (EAC) cell bearing in Swiss albino mice.**Results:** EAU showed the highest antioxidant capacity as (175.80 ± 0.41) mg/g, IC₅₀ value of DPPH scavenging activity was (38.84 ± 0.02) µg/mL and also exhibited maximum lipid peroxidation inhibition activity with the IC₅₀ value of (42.52 ± 0.32) µg/mL than other fractions. The results demonstrate that reducing power of the extract was concentration dependent. In addition, EAU was administered at 50, 100 and 200 mg/kg body weight respectively to EAC cell bearing mice and a significant (*P* < 0.05) decrease in tumor volume, packed cell volume and viable cell count and also increased the life span (17.52%, 42.53% and 62.05%). Hematological profiles were restored to normal levels in MAU treated mice as compared to EAC control mice.**Conclusions:** The results were found to be significant and confirmed that the A. *undulatus* has remarkable antitumor activity with antioxidant potential.

1. Introduction

There are numerous types of oxygen-centered free radicals and other reactive oxygen species (ROS) may produce through various physiological and biochemical processes in the human body. In contrast, at high concentrations of such free radicals can cause oxidative damage to cell structures, including lipids and membranes, proteins and nucleic acids; this damage is often

referred as “oxidative stress”, eventually leading to many chronic diseases, such as atherosclerosis, cancer, diabetes, aging, and other degenerative diseases in humans [1–3]. In addition, free radicals also can cause DNA damage, which can lead to mutations in crucial genes, thus ultimately leading to cancer [4]. It has been reported that ROS are involved in cancer initiation and promotion and malondialdehyde concentration is increased in patients with neoplasms [5]. Moreover, antioxidants that block the activity of free radical damage and suggested that a combination of antioxidants is a powerful adjunctive preventive treatment for cancer [6].

In recent years, the major important group of human medicines still contains phytochemicals being widely used in the world now-a-days, consider for about 50% of the total pharmaceutical market [7]. At the present time, many countries,

*Corresponding author: Md. Reyazul Islam, Department of Bioactive Materials Science, Chonbuk National University, 567 Baekje-daero, Deokjin-gu, Jeonju-si, Jeollabuk-do, Republic of Korea.

Tel: +82 10 3140 1237

E-mails: imdreyazul@gmail.com, imdreyazul@postech.ac.kr

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⁵ These authors contributed equally.

researchers have been extracted and investigated powerful and nontoxic antioxidants from natural sources, especially edible or medicinal plants to prevent the free radical related human disorders, which replace the synthetic compounds, because of their probable carcinogenic activity which are harmful to the lungs and liver [8]. Since ancient times, thousands of species have been known in Bangladesh to have medicinal value and the use of different parts of several medicinal plants for the prevention and treatment of complex diseases.

Aponogeton undulatus (*A. undulatus*) is belongs to the family “Aponogetonaceae” and occurs in India, Sri Lanka, Myanmar, Bangladesh and China [9]. The rootstock of the plant is an important food can be exceptionally useful as a nutrient supplement in many areas where purchasing power is limited because of low incomes. The nutrient composition of the rootstock of *A. undulatus* shows that it can provide an adequate supply of carbohydrates (42.8 g/100 g), protein (8.3 g/100 g), fats (0.7 g/100 g), iron (18.2 g/100 g), calcium (37.2 g/100 g) and some minerals [10]. The literature review exposed that the leaf pastes are used with hot water to treat cuts and wounds and in Ayurveda, the plant is claimed to be effective against cough, tuberculosis, acne, cancer, diarrhea, dysentery, jaundice etc [11]. As reported previously, the primary constituents of *A. undulatus* are tanin and alkaloid which possesses thromolytic activity along with a broad spectrum antibacterial and toxic potentiality [12]. The present study was aimed to evaluate the *in vitro* antioxidants and *in vivo* antitumor activity of *A. undulatus* leaves extracts along with its various organic fractions. To our knowledge, this is the first report to demonstrate *in vivo* antitumor activity of *A. undulatus* extract against Ehrlich ascites carcinoma (EAC) cell in mice.

2. Materials and methods

2.1. Plant material

The plant *A. undulatus* leaves was collected from the village Kamatpara under Panchagarh thana of Panchagarh district, Bangladesh during the month of January 2010 and was identified by the taxonomist of the National Herbarium of Bangladesh, Mirpur, Dhaka whose voucher specimen no. 32095 is maintained in our laboratory for future reference.

2.2. Chemicals and reagents

Folin-chiocaltu phenol reagent was purchased from E. Merck (Germany). 1,1-diphenyl- 2-picryl-hydrazyl (DPPH), ascorbic acid, quercetin, and 5,5'-dithiobis (2-nitrobenzoic acid) (DTNB), L-penicillamine (L-2-amino-3-mercapto-3-methylbutanoic acid), diethylene triamine pentaacetic acid, potassium ferricyanide were purchased from Sigma Co. (St. Louis, MO, U.S.A.). 6-Hydroxy-2,5,7,8-tetramethylchroman-2-carboxylic acid (Trolox) was purchased from Aldrich Chemical Co. (Milwaukee, WI, U.S.A.), Bovine serum albumin and Bleomycin were purchased from Sigma (St. Louis, MO, USA), trichloroacetic acid (TCA) was acquired from Merck (Mumbai, India), thiobarbituric acid (TBA) and nitroblue tetrazolium chloride were purchased from Loba Chemie (Mumbai, India) and DTNB, phenazonium methosulfate and nicotinamide adenine dinucleotide were bought from SISCO (Mumbai, India). All other chemicals and reagents were used of the highest analytical grade.

2.3. Preparation of crude plant extract

The plant material was shade-dried with occasional shifting and then ground to a powder using a mechanical grinder, passed through a sieve No. 40 and stored in an air-tight container. About 500 g of dried powdered plant materials were taken in an amber colored extraction bottle (2.5 L capacity) and the materials were soaked with methanol (1L × 3 times). The sealed bottle was kept for 7 days with occasional shaking and stirring. The combined extracts were filtered through cotton and then Whatman No.1 filter papers and were concentrated with a rotary evaporator under reduced pressure at 45 °C to afford 40 g crude methanolic extract of *A. undulatus* leaves (MAU). The extract was then fractionated by chloroform, ethyl acetate and finally with water to obtain chloroform fraction (CAU, 8.28 g), ethyl acetate fraction (EAU, 7.68 g) and aqueous fraction (WAU, 10.50 g).

2.4. In vitro antioxidant activity

2.4.1. Determination of total phenolic compounds, flavonoids and flavonols

Total phenolic, flavonoid and flavonol contents were determined according to the Cheung method [13] with slight modifications. 1.5 mL sample were mixed with 0.5 mL of Folin-ciocalteu phenol reagent. After 5 min, 1 mL of 10% (w/v) Na₂CO₃ was added to each reaction mixture and reactions were performed 30 min in the dark condition. Absorbance at 725 nm was then recorded using an ultraviolet 1800 spectrophotometer (Shimadzu, Tokyo, Japan). Results were expressed in terms of galic acid equivalents (GAE, mg/g of each extract)). Flavonoids and flavonols content were measured using a colorimetric assay developed previously [14]. Absorbance was measured at 510 nm against a blank, and flavonoids and flavonols content were expressed in terms quercetin equivalent (QAE, mg/g of each extract). All analyses were performed in at least triplicate.

2.4.2. Determination of total antioxidant capacity

Total antioxidant capacity (TAC) of samples/standard was determined by the method reported by Prieto *et al.* [15] with partial modifications. 0.5 mL of samples/standard at different concentrations was mixed with 3 mL of reaction mixture containing 0.6 M sulphuric acid, 28 mM sodium phosphate and 1% ammonium molybdate into the test tubes. The test tubes were incubated at 95 °C for 10 min to complete the reaction. After cooling at RT, the absorbance was measured at 695 nm using a spectrophotometer (Shimadzu, Tokyo, Japan) against blank. Ascorbic acid was used as standard. A typical blank solution contained 3 mL of reaction mixture and the appropriate volume of the same solvent used for the samples/standard were incubated at 95 °C for 10 min and the absorbance was measured at 695 nm. Increased absorbance of the reaction mixture indicated increase total antioxidant capacity of *A. undulatus*.

2.4.3. DPPH free radical scavenging assay

A solution of 0.1 mM DPPH in methanol was prepared and 2.4 mL of this solution was mixed with 1.6 mL of extractives in methanol at different concentrations. The reaction mixture was vortexed thoroughly and left in the dark at RT for 30 min. The

absorbance of the mixture was measured spectrophotometrically at 517 nm. Ascorbic acid was used as reference standard. Percentage DPPH radical scavenging activity was calculated by the following equation:

$$(\% \text{DPPH radical scavenging activity}) = [(A_0 - A_1)/A_0] \times 100$$

Where A_0 is the absorbance of the control, and A_1 is the absorbance of the extractives/standard. Then % of inhibition was plotted against concentration, and from the graph IC_{50} was calculated [16].

2.4.4. Lipid peroxidation inhibition assay

The lipid peroxidation inhibition assay was determined according to the method described by Liu *et al.* [17] with a slight modification. Briefly, excised rat liver was homogenized in buffer and then centrifuged at 13 000 rpm for 15 min to obtain liposome. 0.5 mL of supernatant, 100 μ L 10 mM $FeSO_4$, 100 μ L 0.1 mM AA and 0.3 mL of extractives/standard at different concentrations were mixed together to make the final volume of 1 mL. The reaction mixture was incubated at 37 °C for 20 min. One mL of (28%) TCA and 1.5 mL of (1%) TBA was added immediately after heating. Finally, the reaction mixture was again heated at 100 °C for 15 min and cooled at RT. After cooling, the absorbance was measured at 532 nm. Percentage inhibition of lipid peroxidation was calculated by the following equation:

$$\% \text{ lipid peroxidation inhibition} = [(A_0 - A_1)/A_0] \times 100$$

Where A_0 is the absorbance of the control, and A_1 is the absorbance of the extractives/standard. Then % of inhibition was plotted against concentration, and IC_{50} was calculated from the graph.

2.4.5. Determination of ferrous reducing antioxidant capacity

The reducing power activity of *A. undulatus* leaves extract was determined according to the previously described by Oyaizu *et al.* [18]. Briefly, 0.25 mL samples/standard solution at different concentrations, 0.625 mL of potassium buffer (0.2 M) and 0.625 mL of 1% potassium ferricyanide [$K_3Fe(CN)_6$] solution were added into the test tubes. The reaction mixture was incubated for 20 min at 50 °C to complete the reaction. Then 0.625 mL of 10% TCA solution was added into the test tubes. The total mixture was centrifuged at 3 000 rpm for 10 min. After which, 1.8 mL supernatant was withdrawn from the test tubes and was mixed with 1.8 mL of distilled water and 0.36 mL of 0.1% $FeCl_3$ solution. The absorbance of the solution was measured at 700 nm using a spectrophotometer (Shimadzu, Tokyo, Japan) against blank. A typical blank solution contained the same solution mixture without plant extracts/standard was also incubated under the same condition, and the absorbance of the blank solution was measured at 700 nm. Increased absorbance of the reaction mixture indicates increase reducing capacity.

2.5. In vivo antitumor activity

2.5.1. Animals

Swiss albino mice (25–30 g) of both sexes, purchased from the animal research branch of International Center for Diarrhoeal Diseases and Research, Bangladesh (ICDDR, B) were divided

into six groups, each consisting of twelve animals and used to assess the biological activity. The animals were maintained under standard laboratory conditions [temperature (25 $\pm$ 2) °C] with dark/light cycle (14/10 h) and they were access to standard dry pellet diet and water *ad libitum*. The animals were allowed to acclimatize to the environment for 7 days prior to the experimental session. Animal treatment and maintenance for acute toxicity and anticancer effects were strictly followed in accordance with the Principle of Laboratory Animal Care [19] and the Animal Care and Use Guidelines of Atish Dipankar University of Science & Technology, Dhaka, Bangladesh.

2.5.2. Acute toxicity study

An acute oral toxicity assay was performed in healthy nulliparous and non-pregnant adult female Swiss albino mice (25–30 g) divided into different groups. The test was performed using increasing oral doses of EAU in distilled water (50, 100, 200, 500, and 1 000 mg/kg body weight) that were administered at 20 mL/kg to different test groups. The normal group received distilled water. Following treatment, mice were allowed to feed *ad libitum* and observed for 48 h for any mortality or behavioral changes [20].

2.5.3. Transplantation of tumor

EAC cells were obtained from the Indian Institute of Chemical Biology, Calcutta, India. The EAC cells were maintained *in vivo* in Swiss albino mice by intraperitoneal transplantation of 2×10^6 cells per mouse every 10 days. Ascitic fluid was drawn from EAC tumor bearing mice at the log phase (days 7–8 of tumor bearing) of the tumor cells and each test animal received 0.1 mL of tumor cell suspension containing 2×10^6 tumor cells intraperitoneally [21].

2.5.4. Treatment schedule

The animals were divided into six groups ($n = 12$) and provided with food and water *ad libitum*. All animals in each group received EAC cells (2×10^6 cells/mouse *ip.*) except Group-I served as normal saline control (5 mL/kg body weight *ip.*). Group-II served as EAC control. Twenty four hours after EAC transplantation, Group animals received EAC control (2×10^6 cell/mouse), Group V received bleomycin 0.3 mg/kg body weight (Positive control), groups II, III, and IV, were treated with 50, 100 and 200 mg/kg body weight (*po.*) of the EAU, respectively for nine consecutive days [22]. At 24 h after the last dose was administered, animals were fasted for 18 h, at which point six animals in each group were sacrificed by cardiac puncture for estimation of hematological parameters, as well as to measure antitumor activity and the rest were provided with food and water *ad libitum* and observed to determine if there were any changes in lifespan. The antitumor activity of extract was measured in EAC animals as described below.

2.5.5. Determination of tumor and packed cell volume

Mice were dissected and ascetic fluid was collected from the peritoneal cavity. The volume was measured using a graduated conical centrifuge tube, after which the packed cell volume (PCV) was determined by centrifuging the fluid at 1 000 g for 5 min.

2.5.6. Viable and nonviable tumor cell count

The ascetic fluid was collected in a white blood cell (WBC) pipette and diluted 100 times. A drop of the diluted suspension

was then placed on a Neubauer counting chamber and the cells were stained with Trypan blue (0.4% in normal saline). Cells that did not take up the dye were considered viable, while those that did were considered non-viable. These viable and non-viable cells were then counted.

$$\text{Cell count} = (\text{number of cells} \times \text{dilution factor}) / (\text{area} \times \text{thickness of liquid film})$$

2.5.7. Determination of median survival time and percentage increase in life span

The mortality was monitored by recording the percentage increase in life span (% ILS) and median survival time (MST) according to the following formula [23]:

$$\% \text{ILS} = \left(\frac{\text{Mean survival time of the treated group}}{\text{Mean survival time of the control group}} - 1 \right) \times 100$$

where mean survival time* = (day of first death + day of last death)/2 (*Time is denoted by number of days).

2.5.8. Estimation of hematological parameters

Collected blood was used to estimate the hemoglobin (Hb) content, red blood cell (RBC) and WBC count [22]. Differential counts of WBC were carried out from Leishman stained blood smears [23].

2.6. Statistical analysis

All values were expressed as the mean $\pm$ SEM of three replicate experiments. The analysis was performed by using SPSS statistical package for WINDOWS (version 16.0; SPSS Inc, Chicago). Results related to the reducing power activities were statistically analyzed by applying the Student *t*-test and $P < 0.001$ were considered to be statistically significant. All *in vivo* data are subjected to ANOVA followed by Dunnett's test and $P < 0.05$ were considered to be statistically significant.

3. Result

3.1. In vitro antioxidant activity

3.1.1. Total phenolic, flavonoid and flavonol contents

The total extractable phenolic fractions content of the *A. undulatus* was found to be 95.24 ± 0.42 (MAU), 24.57 ± 1.02 (CAU), 134.29 ± 1.12 (EAU) and 41.29 ± 0.22 (WAU) respectively, which was expressed as galic acid equivalents (GAE, mg/g of each extract). The flavonoid content in the fractions was found 129.40 ± 0.21 (MAU), 28.40 ± 0.61 (CAU), 205.40 ± 0.31 (EAU), and 70.20 ± 0.41 (WAU) respectively, which was expressed as quercetin equivalents (QAE, mg/g of each extract) and total flavonol contents were 46.40 ± 0.11 (MAU), 71.47 ± 0.31 (CAU), 57.80 ± 0.51 (EAU), and 16.00 ± 0.21 (WAU), respectively and fractions was expressed in terms of quercetin equivalent (QAE, mg/g of each extract).

3.1.2. Total antioxidant capacity

The total antioxidant capacity of various organic soluble fractions of *A. undulatus* leaves extract were expressed as the number of equivalents of ascorbic acid. Total antioxidant capacity of EAU fraction showed the highest and was found to be

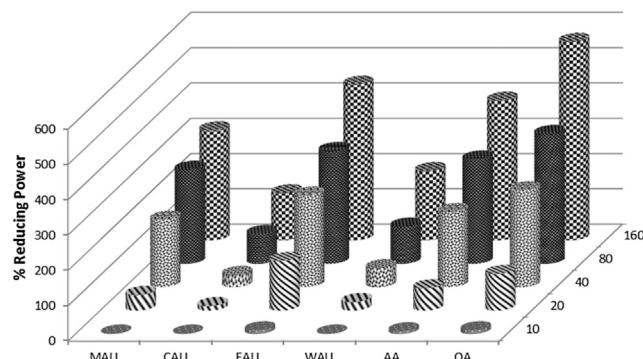


Figure 1. Reducing power activity of *Aponogeton undulatus* leaves extract and its various organic fractions.

The commercial AA: Ascorbic acid, QA: Quercetin used as a standard for comparison.

(175.80 ± 0.41) mg/g followed by MAU, WAU and CAU, with antioxidant capacity of (109.20 ± 0.41), (54.20 ± 0.81), and (32.80 ± 0.11) mg/g equivalent of ascorbic acid, respectively.

3.1.3. DPPH free radical scavenging assay

All the fractions of *A. undulatus* demonstrated H-donor activity. EAU showed the highest DPPH scavenging activity with the IC_{50} value of (38.84 ± 0.02) $\mu\text{g/mL}$ followed by MAU and WAU with the IC_{50} value of (75.24 ± 0.56) and (137.63 ± 0.12) $\mu\text{g/mL}$, respectively. CAU has no activity within the experimental concentration range.

3.1.4. Lipid peroxidation inhibition assay

The lipid peroxides scavenging activity of all the fractions of *A. undulatus* was investigated and compared with standard

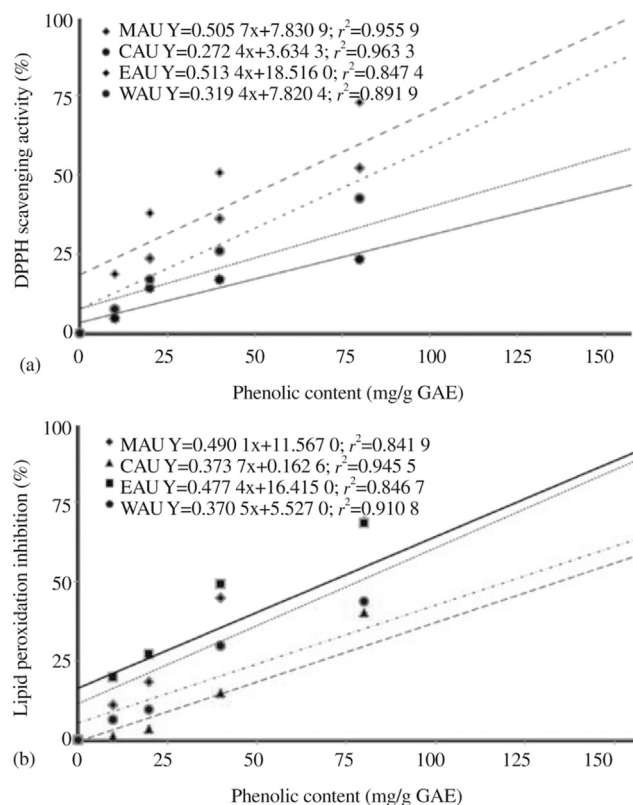


Figure 2. Correlation between phenolic content of the extracts with DPPH scavenging activity (a) and lipid peroxidation inhibition (b).

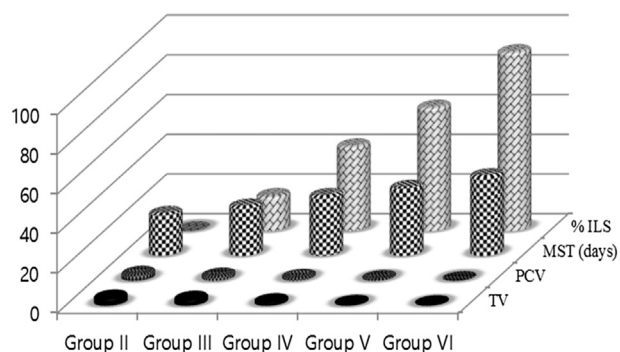


Figure 3. Effects of *Aponogeton undulatus* leaves extracts on tumor volume (TV), PCV, MST, and percentage increase in life span (% ILS) in EAC bearing mice.

$n = 6$ mice per group, * $P < 0.05$ statistically significant when compared with the EAC control group, Group II animals received EAC control (2×10^6 cell/mouse), Group VI received bleomycin 0.3 mg/kg body weight, groups III, IV and V, were treated with 50, 100 and 200 mg/kg body weight (*po.*) of EAU, respectively.

Table 1

Effect of MAU on hematological parameters in EAC bearing mice.

Groups	Hb content (g%)	RBC (cells $\times 10^6/\text{mm}^3$)	WBC (cells $\times 10^6/\text{mm}^3$)
Normal saline (5 mL/kg)	11.52 $\pm$ 0.76	5.22 $\pm$ 0.19	4.23 $\pm$ 0.21
EAC control (2×10^6 cell/mouse)	4.31 $\pm$ 0.56 [#]	2.11 $\pm$ 0.26 [#]	7.02 $\pm$ 0.23 [#]
EAC + MAU (25 mg/kg)	5.12 $\pm$ 0.31*	2.42 $\pm$ 0.14*	6.33 $\pm$ 0.25*
EAC + MAU (50 mg/kg)	6.91 $\pm$ 0.19*	3.22 $\pm$ 0.19*	5.63 $\pm$ 0.21*
EAC + MAU (100 mg/kg)	8.83 $\pm$ 0.23*	4.52 $\pm$ 0.22*	4.93 $\pm$ 0.39*
Bleomycin (0.3 mg/kg)	10.82 $\pm$ 0.61*	4.92 $\pm$ 0.17*	4.51 $\pm$ 0.24*

Each point represents the mean $\pm$ SEM from triplicate determinations ($n = 6$ mice per group), [#] $P < 0.05$ statistically significant when compared with normal saline group, * $P < 0.05$ statistically significant when compared with EAC control group.

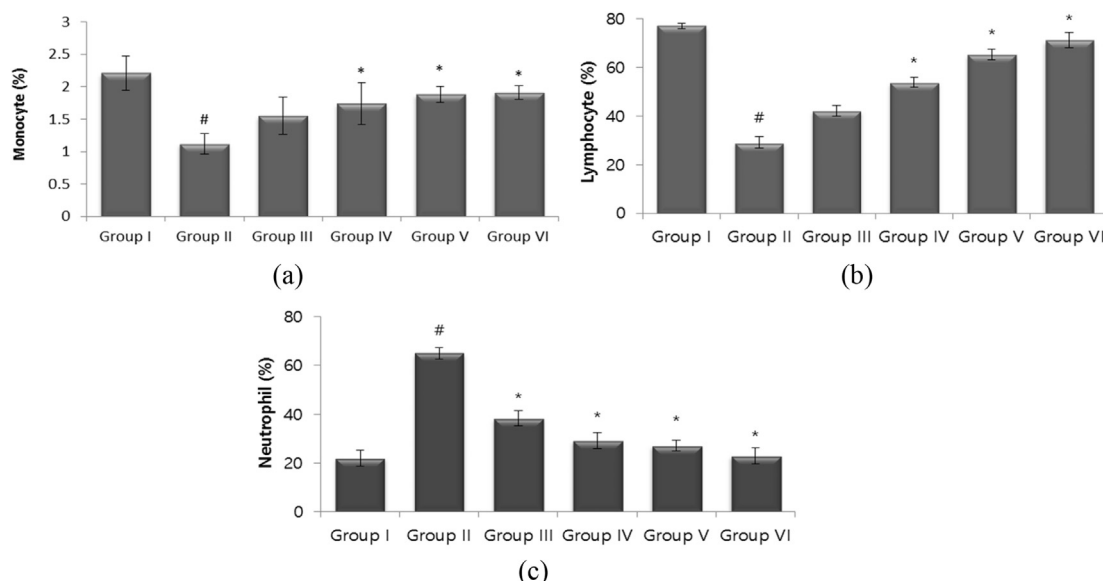


Figure 4. A classical feature of *Aponogeton undulatus* leaves extracts on differential count on WBC, monocyte (a); lymphocyte (b); and neutrophil (c). $n = 6$ mice per group, [#] $P < 0.05$ statistically significant when compared with normal saline group, * $P < 0.05$ statistically significant when compared with EAC control group. Group I animals received normal saline (5 mL/kg) whereas group II received EAC control (2×10^6 cell/mouse), Group VI received bleomycin 0.3 mg/kg body weight, groups III, IV and V, were treated with 25, 50, and 100 mg/kg body weight (*po.*) of MAU, respectively.

ascorbic acid. IC_{50} value of MAU, CAU, EAU and WAU were (49.11 ± 0.41), (138.38 ± 0.61), (42.52 ± 0.32), and (113.98 ± 0.33) $\mu\text{g/mL}$, respectively. On the other hand, the IC_{50} value of standard ascorbic acid was (38.52 ± 0.12) $\mu\text{g/mL}$.

3.1.5. Ferrous reducing antioxidant capacity

In order to measurement of the reductive ability, transformation of Fe^{3+} to Fe^{2+} was investigated in the presence of *A. undulatus* leaves extract and its various organic fractions. Figure 1 shows the reductive capabilities of the *A. undulatus* leaves extract while compared with commercial ascorbic acid and quercetin and the extracts showed dose dependent activity.

There was a clear association between extract phenolic content and DPPH radical scavenging activity linear correlation coefficient (Figure 2a). The results of this investigation showed that the extracts inhibited lipid peroxidation in a dose dependent manner (data not shown) and had a strong and highly significant correlation between total phenol content and lipid peroxidation (Figure 2b).

3.2. In vivo antitumor activity

3.2.1. Acute toxicity studies

Acute toxicity studies mainly aims at establishing the therapeutic index, *i.e.*, the ratio between the pharmacologically effective dose and lethal dose against the same strain and species. MAU leaves was safe at doses as high as 1 000 mg/kg (*po.*) body weight. Behavior of the animals was closely observed for the first 3 h then at the interval of every 4 h during the next 48 h. The extract did not cause mortality, behavioral changes, locomotor ataxia, and diarrhea or weight loss in mice during 48 h of observation. Moreover, food and water intake did not differ among the groups studied.

3.2.2. Tumor growth and survival parameters

EAU 200 mg/kg body weight significantly reduced tumor volume, packed cell volume (Figure 3) and viable tumor cell count, while they increased non-viable tumor cell count (data not shown) relative to the EAC control group. The MST increased to

24.81 ± 0.11 (% ILS = 17.52), 30.09 ± 0.17 (% ILS = 42.53) and 34.21 ± 0.11 (% ILS = 62.05) upon administration of EAU at doses of 50, 100 and 200 mg/kg body weight, respectively, while the EAC control (2×10^6 cell/mouse) and the reference drug bleomycin showed survival times of 21.11 ± 0.12 and 40.11 ± 0.11 (% ILS = 90.00), respectively (Figure 3). Finally the change of body weight (data not shown) of the animal suggested that EAU had the potential to inhibit tumor growth.

3.2.3. Hematological parameters

Hematological parameters of tumor bearing mice were found to be significantly altered relative to the normal group (Table 1). The total WBC count was found to be increased ($P < 0.05$) with a reduction of Hb content and RBC count in EAC control animals when compared with the normal saline group. Additionally, monocytes (Figure 4a) and lymphocytes (Figure 4b) were observed to decrease, while neutrophils (Figure 4c) increased in the EAC control group when compared with the normal saline group. Treatment with 200 mg/kg body weight (*po.*) of MAU significantly changes these altered parameters to near normal values (Figure 4).

4. Discussion

Plants contain free radical scavengers like phenolic compounds, flavonoids and flavonols which are responsible for such antioxidants and antitumor activities that could be isolated and then used as antioxidants for the prevention and treatment of free radical-related disorders [24,25]. In this research, we studied MAU along with its various organic fractions as well as the *in vitro* antioxidant and *in vivo* antitumor activity against EAC cell bearing in Swiss albino mice.

Phenolic compounds are considered as secondary metabolites and these phytochemical compounds derived from phenylalanine and tyrosine occur ubiquitously in plants and are diversified [26]. The highest total phenolic contents exhibited from methanol extract, whereas the contents obtained with residual aqueous fraction were much smaller that is in agreement with other reports [27]. However, flavonoids are often found to be glycosylated form of polyphenolic compounds with different molecular structure in plants. But in many plants flavonoids can also present as aglycone form. Glycosylated flavonoids are soluble with polar solvents such as methanol and aglycone flavonoids are soluble with non polar solvent such as hexane or less polar solvents such as toluene [28]. In this study, *A. undulatus* leaves of phenols, flavonoids and flavonols present as glycosylated form, because of the presence of hydroxyl groups in flavonoids extraction done by alcoholic solvents like methyl alcohol. Then based on the capacity of antioxidant and free radical scavenging activities of flavonoids the solution was further extracted by ethyl acetate. Depends upon the position of hydroxyl groups on the phenyl ring whereas for the significant characteristics of molecular structure is very much soluble with polar organic solvents. In our findings, the best performance of antioxidant activity is carried out by EAU as (175.80 ± 0.41) mg/g plant extract (in ASC) of *A. undulatus*. The total content of phenols, flavonoids and flavonols of *A. undulatus* in the fractions were isolated to the order of EAU > MAU > WAU > CAU.

EAU fraction showed the highest DPPH scavenging effects and lipid peroxidation inhibition as a dose dependent manner

(data not shown) with the IC_{50} value of (38.84 ± 0.02) $\mu\text{g/mL}$ and (42.52 ± 0.32) $\mu\text{g/mL}$, respectively where the standard ascorbic acid activity found to be IC_{50} (18.84 ± 0.02) $\mu\text{g/mL}$ and IC_{50} (38.52 ± 0.12) $\mu\text{g/mL}$, respectively. There was a clear association between extract phenolic content and DPPH radical scavenging activity linear correlation coefficient. Peroxidation of lipid is a natural phenomenon and occurs on its exposure to oxygen. Recently, free radicals-induced lipid peroxidation has gained much importance because of its involvement in several pathological conditions such as aging, wound healing, oxygen toxicity, liver disorders, inflammations *etc.* The results of this investigation showed that the extracts inhibited lipid peroxidation in a dose dependent manner (data not shown) and had a strong and highly significant correlation between total phenol content and lipid peroxidation. Phenolic are commonly found in fruits and have been reported to exhibit antioxidant and to scavenge free radicals through hydrogen or electron donation [29]. Previous studies identified a positive relationship between phenolic functional groups and DPPH free radical scavenging activity [30].

Initially reported as a spontaneous murine mammary adenocarcinoma, the Ehrlich tumor can be grown in the majority of mouse strains and is accepted as a transplantable tumor model to examine the antitumor effects of a number of substances [31]. Ehrlich ascites carcinoma, B-cell lymph proliferative diseases, chronic lymphocytic leukemia diseases perform their mechanism by uncontrolled blood cell regulation. They need to treat by kinase inhibitor enzyme. Kinase inhibitors binds with kinase protein whereas flavonoids with base nucleosides in presence of nitrogenous base it form intermolecular bond with target protein kinase and stop cell further growth [22]. A rapid increase in ascetic tumor volume was observed in EAC tumor-bearing mice and the treatment with *A. undulatus* extracts reduced the intraperitoneal tumor burden, thereby reducing the tumor volume, tumor weight and viable tumor cell count, while increasing the life span of the tumor-bearing mice. Therefore, it may be hypothesized that the increase in lifespan of EAC-bearing mice in response to EAU at 200 mg/kg may be due to a decrease in nutritional fluid volume and a delay in cell division, similar result has been previously reported [23]. Reductions in viable cell count and increased non-viable cell count towards normal in tumor hosts indicate antitumor effects against EAC cells in mice. These results indicate that EAU have a direct association with tumor cells at higher doses as they absorb the anticancer drug by direct absorption in the peritoneal cavity, resulting in lysis of the cells via a direct and cytotoxic mechanism. Anemia and myelo-suppression have frequently been observed in ascites carcinoma due to iron insufficiency, either by hemolytic or myelopathic conditions, which finally lead to reduced numbers of RBC [32]. Treatment with 100 mg/kg body weight (*po.*) of MAU returned the Hb content, RBC and WBC count to more or less normal levels, supporting its hematopoietic protecting activity without inducing myelotoxicity, which is the most common side effect of cancer chemotherapy.

Plant-derived extracts with antioxidant potential have demonstrated cytotoxicity against tumor cells and antitumor activity in experimental animals [33]. The cytotoxic and antitumor activity of plant-derived products occurs either through the induction of apoptosis or the inhibition of neovascularization [34]. In the present study, higher doses of EAU dramatically reduced tumor growth and the viability of the tumor cells and normalized the hematological profiles, increasing the life span compared with the EAC control mice. The decrease in lipid

peroxidation and free radical scavenging effects and reducing power activity in EAU indicated the potential of *A. undulatus* extract as an inhibitor of intracellular free radical induced by oxidative stress.

In conclusion, the results of present study indicate that the *A. undulatus* leaves extract along with its various organic fractions have promising antioxidant properties and remarkable antitumor activity as observed in *in vivo* studies. Our prospect goal is to investigate the isolation and characterization of lead compounds accountable for the above mentioned activity of this plant and find out the ideal conditions to preserve *A. undulatus* samples so that there would be little or no damage to its antioxidant and biological properties.

Conflict of interest statement

We declare that we have no conflicts of interest.

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